The Utah Teapot in CONV Geometry: Nodes on the Surface, Rays by Transport

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Abstract—We rebuild the Utah Teapot from points that lie on its own surface. The representation is a lattice of nodes reconstructed by Compact Ordered Nodal Variation (CONV) with ledger-free admission. Three additions make it a geometry and not only an image filter. Declared seams with cubic-exact one-sided stencils let the lattice reproduce bicubic Bézier patches exactly: with four node intervals per patch, 560 on-surface nodes reproduce the 28 original patches to 2×10^{-15}. A symmetric bubble limiter replaces the standard one, which is discontinuous in the data and can open cracks in a surface; the new limiter is continuous and keeps the variation bound. And because a CONV cell depends on a fixed block of nodes, the surface has a pointwise evaluator with exact derivatives. We render it by correspondence transport, which sends a lattice of known parameters forward to the screen and lets each pixel correct a prediction with a Newton step; it is 1.6 to 2.0 times faster than a per-ray tracer of the same surface and agrees with a fine reference on 99.9% of surface pixels. The representation needs no patches, control points, degrees or knots, only points on a surface, so later versions enter the same way: the 2026 Utah Teapot is rebuilt from 9 by 9 samples of each of its surfaces, interior included. We then measure against the defects that the later teapots were made to fix. A node lattice makes the body 233 times rounder with the same node count and renders without a mesh. Taken patch by patch, the 2026 shape keeps its normals to 0.11° across joints but not its curvature continuity. Taken as whole sheets it does better: the body with its interior is one lattice, the lid another, with rows placed by a piecewise projective warp whose density is set by CONV's own error, and the mean relative curvature jump along the profile falls from 0.16 to 0.003. The handle and spout are joined to the body by blends that interpolate between the two surfaces. These joins are ours, not the 2026 model's chamfers, and we report how closely they meet the body. Given the same nodes on targets native to no method, CONV needs as few as the globally solved C^2 spline and a half to a quarter of what a local bicubic needs; the whole 2026 teapot is 2912 nodes. Finally the surface can be baked: sampled once to vertices chosen by its own error, drawn by a CPU rasterizer with no evaluation per frame, and lit from a texture that the ray tracer paints, with reflections that follow the camera taken from emitters and not traced. The same mesh seeds the ray tracer, which then needs about one surface evaluation per pixel.

Index Terms—Utah Teapot, interpolating surface, variation diminishing, Bézier patch, ray tracing parametric surfaces, correspondence transport.

I. Introduction

The Utah Teapot was modelled by Martin Newell in 1974–75 as 28 bicubic Bézier patches and scaled to its familiar proportions by Jim Blinn in 1976 [1], [2], [3], [4]. Its control points mostly lie off the surface: a Bézier patch passes through only four of its sixteen. Every later version of the model addressed something that the original patches leave open. Crow added a bottom. Driskill made the body round, trimmed the handle and spout to the body and added an interior. Fish improved the interior. Yuksel's 2026 version adds chamfers, a lighter interior, texture layouts and curvature continuity for smooth reflections [5], [6].

This paper puts the teapot in a representation whose every control value is a point of the surface. CONV [8] interpolates samples on a regular lattice by a piecewise quintic whose derivative cannot change sign more often than the samples do. It was built for images. Used for positions, it gives a surface through a lattice of nodes, and a node can be moved, measured, textured or collided with as the surface point it is.

The contributions are these.

1) Seamed CONV lattices (Section III). A seam cuts the difference stencils along a node row or column. With cubic-exact end stencils, a seamed lattice reproduces any piecewise bicubic surface whose patch joints lie on seams, wherever admission leaves the proposal unchanged. For the teapot that is everywhere.
2) A continuous bubble limiter (Section IV). We show that the standard limiter jumps, give a symmetric replacement, and prove that it keeps the variation bound.
3) A pointwise surface evaluator and transport ray tracing (Sections V and X), with a measured comparison against per-ray tracing of the same surface.
4) An account against the later teapots (Section IX): what the representation improves, what it leaves unchanged, and what it does not address.

An engine that implements all of it accompanies the paper and runs in a browser on one CPU thread. Its controls follow the official Utah Teapot page: a choice of version (1975, 1976, 1987, 2026), switches for the handle, spout, lid, body and interior, Blinn's scale, wireframe, texture coordinates per part or per patch, and reflection lines; it adds gloss and a rigid-body drop.

A. For readers of the spline literature

CONV on a lattice is an interpolating surface formulation, and the questions usually put to one have short answers here.

- Interpolation. The surface passes through every node. There are no off-surface controls and no handles.

- Locality and cost. A cell depends on the six samples around it in each direction (eight next to a seam). Every step is closed form, with constant work per cell and no global solve.
- Precision. Linear data are reproduced exactly, so straight pieces need no special constraint. With seams, cubics are reproduced exactly (Proposition 1).
- Continuity. C^2 across node lines wherever the limiter is inactive; C^0 across a seam, which is how a crease is modelled. Curvature is not controlled at a seam, and Section IX measures what that costs.
- Shape. The derivative along a line changes sign no more often than the samples do. Overshoot next to a crease or a step is excluded by construction, not by tuning.
- Dependence on the data. Continuous and piecewise affine in the nodes (Theorem 2).
- Spacing. Nodes need not be evenly spaced on the surface. A sheet is a flat lattice warped onto the shape, and its rows may be placed by a projective warp (Section VII); the interpolation is in the lattice index.
- What it lacks. Coordinates are limited separately, so where the limiter acts the surface is not rotation invariant. One lattice has the topology of a disc, a tube or a sphere; a handle needs a join. It is not a subdivision or a convex-hull scheme.

II. CONV Lines

We restate what is needed from [8]. For samples y_i on a line, with secants $\delta_i = y_{i+1} - y_i$, each node receives a scaled jet (m_i, q_i) from a difference stencil. The cell $[i, i+1]$ gets five raw currents, the differences of six quintic Bézier controls:

$$\begin{aligned} a_{i0} &= \tfrac{m_i}{5}, & a_{i1} &= \tfrac{m_i}{5} + \tfrac{q_i}{20}, \\ a_{i4} &= \tfrac{m_{i+1}}{5}, & a_{i3} &= \tfrac{m_{i+1}}{5} - \tfrac{q_{i+1}}{20}, \end{aligned} \tag{1}$$

and $a_{i2} = \delta_i - a_{i0} - a_{i1} - a_{i3} - a_{i4}$. Ledger-free admission clips a_i into a box built from the two neighbouring secants and terminal currents, and restores the cell mass δ_i by one capped common shift. The admitted currents c_i satisfy $\operatorname{var}(c^\flat) \le \operatorname{var}(\delta)$ for any proposal whose cells sum to their secants; the proof in [8] uses nothing else. The cell is then

$$y(i+t) = y_i + \sum_{j=0}^{4} c_{ij}\, \tau_j(t) + \omega_i\, \psi(t), \tag{2}$$

with $\tau_j = \sum_{r>j} B_r^5$ and the odd bubble $\psi(t) = [t(1-t)]^3(2t-1)$. In the interior of a line the jets are the fourth-order centred stencils, and a cell depends on the six samples $y_{i-2}, \dots, y_{i+3}$ only.

III. Seams and Cubic Precision

A first attempt to put the teapot on a plain CONV lattice fails near its creases and patch joints. With four node intervals per patch the worst deviation from the Bézier surface is 5.9×10^{-2} on a body 4.0 wide, against 1.05×10^{-1} for the same nodes joined bilinearly; next to the crease of a flat bottom cap, CONV is worse than linear. The variation bound holds, and there is no overshoot. The loss is in the jets: a centred stencil that straddles a crease measures a slope that belongs to neither side.

A. Seams

A seam is a node row or column across which no difference stencil may reach. Values remain continuous across it. The two ends of an open line are seams; a closed ring without seams is periodic. A node on a seam has two jets, one from each side. For a segment of at least four nodes between seams we use the stencils that are exact for cubics and still fit the six-sample window:

$$\begin{aligned} m_0 &= \tfrac{1}{6}(-11y_0 + 18y_1 - 9y_2 + 2y_3), \\ q_0 &= 2y_0 - 5y_1 + 4y_2 - y_3, \\ m_1 &= \tfrac{1}{6}(-2y_0 - 3y_1 + 6y_2 - y_3), \\ q_1 &= y_0 - 2y_1 + y_2, \end{aligned} \tag{3}$$

with their mirror images at the other end and the centred stencils from the third node on. Admission is unchanged, except that the terminal current a cell sees in its neighbour across a seam, $a_{i-1,4}$ or $a_{i+1,0}$, is now the neighbour's own one-sided value. This is the case for which the knot-exclusivity terms of the box were written; without seams they collapse to the sign of a shared slope. A cell next to a seam depends on $y_{i-3}, \dots, y_{i+4}$.

Proposition 1 (Cubic precision). Let a segment between two seams have at least four nodes, and let its samples be those of a cubic polynomial at equally spaced parameters. If admission leaves every raw current of the segment unchanged, the CONV line equals that cubic on the segment, with zero bubble.

Proof. The stencils (3) and the centred stencils return the first and second derivatives of a cubic exactly. The raw quintic of each cell is then the quintic Hermite interpolant of exact two-jets of a cubic at both ends, which is that cubic. With $c_i = a_i$ the second term of the bubble fit vanishes, and its first term is a fifth difference, which vanishes on a cubic. □

Theorem 1 (Exact Bézier reproduction). Let a surface be a grid of bicubic Bézier patches, sampled at $n \ge 3$ equal parameter intervals per patch in each direction, with seams on every patch boundary. If admission is inactive in both factors, the CONV surface of Section V equals the Bézier surface.

Proof. Along a node row inside one patch the samples are those of a cubic, so by Proposition 1 every row line equals the surface along that row at every u. The second factor then interpolates, at fixed u, values that are samples of the cubic $v \mapsto S(u, v)$ within the patch. □

The hypothesis is checkable, and for the teapot it holds. Table I gives the measured deviation at eight samples per node interval over all four parts. From four intervals per patch on, the teapot's own geometry passes admission

untouched and the two surfaces agree to rounding. At three intervals a few cells are clipped. At two there are too few nodes for a cubic stencil.

TABLE I
Deviation from the Bézier teapot (body 4.0 wide)

		Seamed, cubic ends		Seamless	
n	Nodes	max	rms	max	rms
2	152	2.3×10^{-1}	5.4×10^{-2}	2.1×10^{-1}	4.8×10^{-2}
3	324	2.9×10^{-5}	1.3×10^{-6}	8.8×10^{-2}	1.7×10^{-2}
4	560	2.3×10^{-15}	4.1×10^{-16}	5.9×10^{-2}	9.5×10^{-3}
6	1224	3.4×10^{-15}	6.2×10^{-16}	3.5×10^{-2}	4.7×10^{-3}
8	2144	2.4×10^{-15}	4.1×10^{-16}	2.5×10^{-2}	2.9×10^{-3}

The node count is not smaller than the control-point count; 28 patches have 448 control points and the exact lattice has 560 nodes, 64 of them on a flat bottom cap that is not one of the 28 patches. What changes is where they are. All 560 are points of the teapot.

IV. A Continuous Bubble

The bubble amplitude ω_i is fitted and then limited so that the degree-six Bernstein coefficients of the cell's derivative keep their signs. Writing d_2, d_3, d_4 for the three coefficients the bubble touches, the enriched ones are $d_2 - \omega/5$, $d_3 + 2\omega/5$ and $d_4 - \omega/5$. The standard limiter [8] lets each keep its sign one-sidedly: for $d_2 > 0$ it requires only $\omega \le 5d_2$.

Proposition 2 (The standard limiter is discontinuous). Fix a fitted amplitude $\widehat{\omega} > 0$ and let d_2 pass through zero with d_3, d_4 fixed and nonzero. For $d_2 \downarrow 0$ the admitted amplitude tends to 0. For $d_2 \uparrow 0$ the constraint from d_2 is a lower bound that tends to 0, and the admitted amplitude tends to $\min\{\widehat{\omega}, U\}$, where $U > 0$ is the upper bound from the other two coefficients whenever one exists.

On a line this is a jump in the data-to-value map. On a surface it is a crack: the second factor's inputs vary continuously with u, and where one of its coefficients changes sign the surface position jumps. The defect was observed in an earlier cloth experiment; here is its cause and a cure.

Theorem 2 (Symmetric limiter). Let

$$M_i = \min\{5|d_{i2}|,\ \tfrac{5}{2}|d_{i3}|,\ 5|d_{i4}|\},$$
$$\omega_i = \max\{-M_i, \min\{M_i, \widehat{\omega}_i\}\}. \tag{4}$$

Then ω_i is a continuous piecewise-affine function of the cell's samples, no enriched coefficient changes sign, and the variation bound of [8] holds unchanged. Wherever $|\widehat{\omega}_i| \le M_i$ the amplitude equals the fitted one, so the leading-error reduction 1681/1950 is retained.

Proof. $|\omega_i| \le M_i$ gives $|\omega_i/5| \le |d_{i2}|$, $|2\omega_i/5| \le |d_{i3}|$ and $|\omega_i/5| \le |d_{i4}|$, so each enriched coefficient is zero or has the sign of the one it replaces. That is the only property of the amplitude used in the variation proof. $\widehat{\omega}_i$ is linear in the samples and the admitted currents, which are continuous and piecewise affine; M_i is a minimum of absolute values of linear forms of them; and a clamp is continuous in its three arguments. □

The symmetric bound is the tighter of the two, since it limits both signs of the amplitude. On resolved smooth data the fit is of fifth order in the spacing and the coefficients of first order, so neither limiter acts and the two agree. Table II measures the difference where it matters: the largest change of position between surface samples one step apart along u, on the teapot body with its nodes displaced by noise of standard deviation 0.08. A continuous surface gives a change proportional to the step. The standard limiter leaves a jump of 1.3×10^{-4} at a step of 10^{-6}, about a hundred times the continuous value.

TABLE II
Largest position change over one parameter step

Bubble	step 10^{-4}	step 10^{-6}
off	1.8×10^{-3}	1.4×10^{-6}
standard	2.0×10^{-3}	1.3×10^{-4}
symmetric	1.9×10^{-3}	1.4×10^{-6}

With the symmetric limiter the whole factor, bubble included, is continuous in the data, and the surface of the next section is watertight within each part.

V. The Surface

A part is a lattice of nodes $P_{kl} \in \mathbb{R}^3$, closed around (u) and open along (v). The surface is the fixed-order composition of two CONV factors, evaluated at a point. For (u, v) in cell (i, j) with local parameters (s, t):

1) for each node row $l = j - 3, \ldots, j + 4$ that exists, evaluate the row's CONV cell i at s, giving $R_l(s)$;
2) admit one CONV cell along v from the window $R_{j-3}, \ldots, R_{j+4}$ and evaluate it at t.

Each coordinate is treated separately in the object's own frame. The admitted currents of step 1 do not depend on s and are compiled once per lattice; step 2 is admitted at every query, because its inputs depend on u. A point costs 24 polynomial evaluations and three admissions. This is exactly the ordered composition of [8], not an approximation of it; the ledger form could not be evaluated this way, since its sign ledger needs the whole line.

Derivatives. Admission is piecewise affine, so forward-mode differentiation through its minima and maxima is exact wherever a derivative exists. We evaluate both steps on dual numbers that carry $\partial/\partial s$ and $\partial/\partial t$ and obtain S_u and S_v with the point. The normal is their cross product.

Regularity. Within a part the surface is continuous (Theorem 2). Where admission and the limiter are inactive it is C^2 across node lines away from seams, because adjacent cells share two-jets and ψ has triple zeros. Across a seam it is C^0 by design.

Invariance. Each factor commutes with translation and with scaling of a coordinate, so an axis-aligned affine map of the nodes maps the surface exactly; Blinn's vertical scaling of Newell's teapot is such a map. Where admission

is inactive the operator is linear and commutes with every affine map. Where it is active, a rotation of the nodes does not in general rotate the surface, since coordinates are admitted separately.

VI. The Teapot

The source is the original data from the Utah Model Repository [7]: 28 patches, without a bottom. They assemble into four parts, each a grid of patches closed around: the body (rim, upper and lower body: four patches around, three along), the lid (knob and lid: four by two), the handle and the spout (two half tubes around, two along). We add an optional flat cap under the body, as Blinn suggested [4]; it meets the body at a 90° crease, which a seam keeps exactly, and the lowest point of the CONV body is the plane itself.

Three lattice styles are provided for this data.

- Exact: seams on every patch boundary, nodes at equal Bézier parameters. This is the 1976 teapot to rounding (Table I).
- Round: as exact along v, but every ring of the body and lid lies on a true circle and is closed without seams.
- Seamless: no seams, the closure of [8]. Smooth across the patch joints, at the deviation of Table I.

A. Other versions, from samples alone

Nothing in Section V asks where the nodes came from. The official page generates meshes of every version, and a mesh vertex of a patch is a point of its surface. We took, for the 2026 version, the 9×9 vertices of each of its 66 untrimmed surfaces at eight intervals per patch, and for the 1987 version the four patches of Crow's round bottom. Symmetry leaves 19 distinct patches of the 66. Each becomes one open lattice with cubic ends. No control point, degree, knot vector or continuity construction of the source model is used or known to us; the samples are not even uniform in the source parameter.

The versions offered are then: 1975, Newell's proportions, which is the 1976 lattice with its heights scaled by 4/3, an exact operation (Section V); 1976; 1987, the 1976 lattice raised by 0.15 with the sampled bottom; and 2026, 5346 nodes on 46 exterior and 20 interior surfaces. Adjacent lattices share their edge samples, and a CONV line along an edge depends on that edge's samples alone, so the lattices meet without gaps: the largest measured gap over 52 joints is 9×10^{-8}, the rounding of the single-precision samples. Fig. 1 shows them.

What is left out is stated plainly. The 2026 model trims the body, handle and spout against each other and adds chamfers at the joins; those surfaces come out of the generator as irregular meshes, not grids, and are not included, so the handle and spout still pass through the wall. The 1992 and 2006 versions, whose difference is in trimmed surfaces and interiors, are not included.

VII. One Sheet per Part, and Joins

The patch decomposition of the previous subsection is inherited from the source, and its seams are where curvature continuity is lost (Section IX). A sheet does not need them. A lattice closed around and open along, with a pole at each end, is a sphere, and that is a bowl with its inside: the body's profile runs from the outer bottom centre up the outside, over the rim, down the inside, to the inner bottom centre. Revolved on true circles it is one sheet. The lid, from the apex of its knob around its edge and back along its underside, is another. The handle is a tube; the spout is a tube that runs out along the outside, around the lip and back down its channel. Four sheets replace 66 patches.

A. Projective row placement

The rim and the foot turn within a few hundredths of a unit; the wall barely turns at all. CONV interpolates in the row index, so what matters is that the profile be a smooth function of that index, not that rows be evenly spaced. We place rows by a piecewise projective warp. Stations $s_0 < \cdots < s_m$ are laid along the profile's arc length, and each is given a wanted spacing h_k, within limits and bounded in its rate of change; how h_k is chosen is the next subsection. Between stations $\sqrt{h}$ is linear. Writing $p = \sqrt{h}$, a span of length d then holds

$$n = \int \frac{ds}{h} = \frac{d}{p_0\, p_1} \tag{5}$$

rows, and the arc length that holds c rows from the span's start is

$$x(c) = \frac{c\, p_0^2}{1 - c\, p_0\, (p_1 - p_0)/d}, \tag{6}$$

a Möbius function of c. A projective map of a uniform grid is exactly a spacing whose square root is linear, so each span is one projective warp and adjacent spans agree in spacing where they meet. One global scale is found by bisection on the row count. A row is placed on each true crease, which is detected from the one-sided tangents of the source curve and receives a seam. The 2026 body has one such crease, of 115° at the rim, and the lid one of 90° at its edge; everything else is seamless.

The source for this is denser than before: 49 points per patch edge along the two meridians, and 17×17 points per patch of the tubes. The earlier 9×9 reconstruction had tangent kinks of up to 1.5° at its own joints, which a sheet fitted to it would inherit.

B. Rows by CONV's own error

Our first choice of h was the profile's radius of curvature. It is a poor one: the 2026 body has three bends of radius about 0.03 (at the foot, the lip and inside the rim) that it under-resolves, while rows pile up at the rim's crease, where the radius measured across the crease is meaninglessly small. The better density comes from the interpolant itself. Through the current rows the meridian is a pair of CONV lines. In each cell we measure its

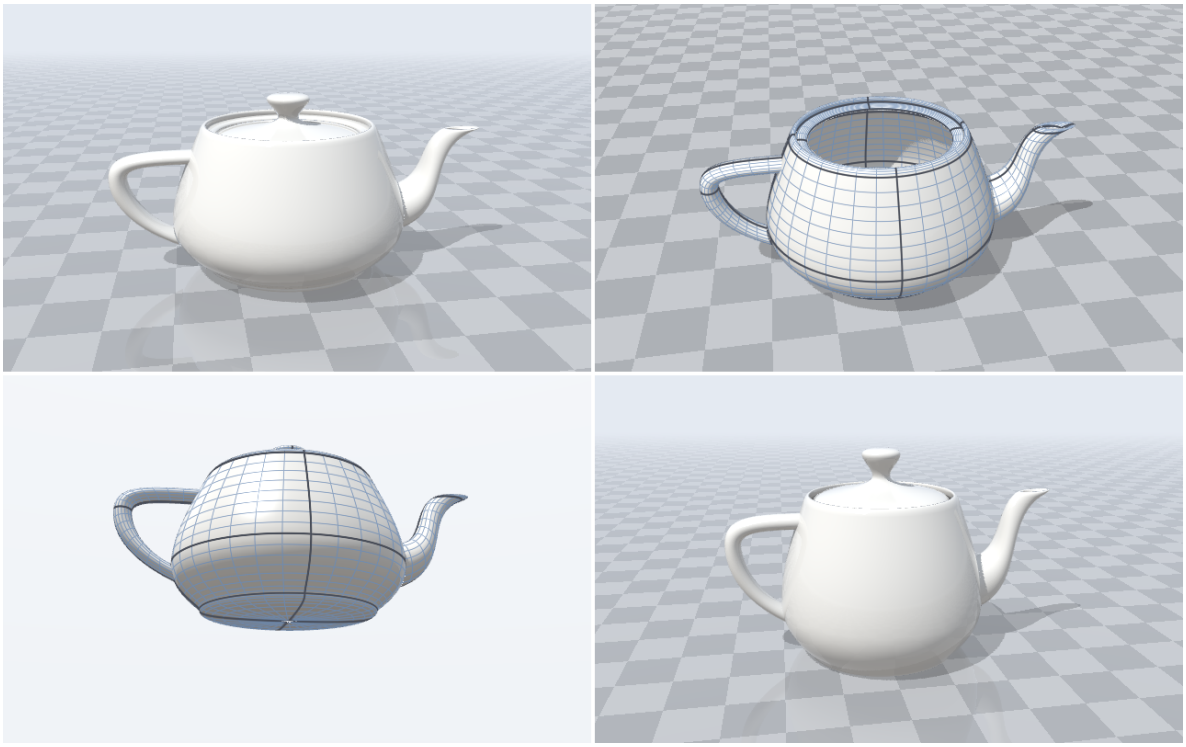


Fig. 1. Versions as node lattices. Top: the 2026 version from surface samples, glazed; the same with the lid removed, showing its interior and the node lines of each 9×9 lattice. Bottom: the 1987 version from below, with Crow's round bottom; the 1975 version in Newell's proportions.

distance e from the source profile and form $g = e/h^4$, which for a fourth-order interpolant estimates how hard the profile is there, independent of the spacing used to measure it. The spacing that makes every cell err alike is $h \propto g^{-1/4}$, the equidistribution rule of de Boor [9]. That h is handed to the projective grading above, the rows are replaced, and the step is repeated eight times with the estimate averaged geometrically between rounds; the plan with the smallest error is kept. The only inputs are the CONV line and the source.

With 16 nodes around, the body's largest distance from its profile falls from 5.8×10^{-3} to 1.0×10^{-3} at 64 rows and from 3.7×10^{-3} to 2.6×10^{-4} at 96, between 6 and 14 times for the same nodes. Rows evenly spaced in arc length are worse again (3.3×10^{-2} at 64 rows).

C. Measurements

Table III gives, for the body and lid sheets, the distance of the sheet's meridian from the source profile and the continuity along it at every row that is not a crease, by one-sided finite differences.

TABLE III
Unified sheets of the 2026 teapot

Sheet	Lattice	Distance max	Distance rms	Tangent turn max	Curv. jump mean rel.
Body	12×73	6.6×10^{-4}	1.8×10^{-4}	0.04°	0.003
Lid	12×55	3.1×10^{-4}	7.6×10^{-5}	0.26°	0.001

Against the per-patch lattices of Table IX, the mean relative curvature jump falls from 0.16 to 0.003 on the body and 0.001 on the lid, with a third of the nodes for those two parts (1536 against 4536). It is not zero, and the row count is a real parameter: with 48 rows the body strays by 3.0×10^{-3} and its normal by up to 21° inside the three tight bends, which then hold too few rows to turn through. Twelve nodes around are enough for the round parts; the tubes, whose sections are not circles, take sixteen.

D. Joins as interpolants

Where a tube passes through the body the two surfaces meet along a closed curve Γ. The join is a third surface that interpolates between them. For a point Γ_k of the curve and $\tau \in [0,1]$ across a band of width w,

$$C(k,\tau) = \big(1-\beta(\tau)\big)\,A_k\big(w(1-\tau)\big) + \beta(\tau)\,B_k(w\tau), \quad (7)$$

where $A_k(d)$ is the point of the body at distance d from Γ_k, away from the tube, $B_k(d)$ the point of the tube at distance d from Γ_k, away from the body, and $\beta(\tau) = \tau^3(10 - 15\tau + 6\tau^2)$. At $\tau = 0$ the join is the body and at $\tau = 1$ it is the tube; since β' and β'' vanish at both ends, the curve $\tau \mapsto C(k,\tau)$ leaves the body and arrives on the tube with their tangent and curvature. On Γ the two surfaces are in unison, so the convex combination is taken between points that approach each other. The buried part of the tube and the covered part of the body are trimmed in their parameter domains, by a cut around the tube and a polygon on the body. The join is stored as its two sides: a lattice of 16 by 3 nodes for A and another

for B, with β evaluated in closed form (described after Table IV). The intersection curve is found by bisection along each column of the tube against the body's radius at that height, which the body's being a surface of revolution makes a one-dimensional search.

Four joins result: the two ends of the handle, the base of the spout, and the spout's channel where it opens into the interior. Where a tube's end only touches the body, as at the lower end of the handle, the join starts at the end of the tube. Table IV measures the band's edge on the body against the body itself. Each join turns the normal through about 90° across its band, which is the crease it replaces.

TABLE IV
Joins: the band's edge against the body

Join	Gap max	Normal mismatch max
Handle, upper	1.1×10^{-7}	0.005°
Handle, lower	1.2×10^{-7}	0.005°
Spout base	1.2×10^{-7}	0.005°
Spout channel, interior	1.1×10^{-7}	0.005°

The band's points are found by walking the given distance along each surface in short steps that read the surface's own speed, not by one step from the speed at the start, which is wrong where rows are graded.

How the band is stored decides how many nodes it needs and how well its edges meet the surfaces. If C itself is sampled as a lattice of rows across the band, CONV takes the tangent at each edge from the first four rows by a one-sided stencil, and over those rows β is already rising: the normal at the edge was off by 3° with nine rows (seventeen on the interior join) and by 1° with seventeen everywhere. But β is known in closed form, and A and B are short walks on the two surfaces. So the band stores where those walks are: for A a lattice of 16 by 3 points of the body's parameter domain, for B one coordinate along the tube at each of 16 by 3 places. At evaluation CONV interpolates the parameters, the body and the tube are evaluated there, and the two are blended with β. Each side is then a piece of the surface it names. At the band's edge the blend weight and its first two derivatives vanish, so the band has the body's position, normal and curvature there by construction; the measured gap of 10^{-7} and mismatch of 0.005° in Table IV are the offset at which they were sampled. An earlier version stored the two sides as points near the surfaces and met the body within 1°, which chrome showed as a line along the join from some angles. The four joins take 384 nodes in place of 704. To whoever samples the surface by its cells the band presents four cells for each interval between its rows, since it turns the normal through 90° across two intervals.

TABLE V
The four joins stored as two sides, against the same with 65 rows a side: largest distance and largest normal deviation inside the band. At the edges the deviation is zero for every row count.

Rows a side	Nodes	Distance	Normal
2	256	2.8×10^{-3}	1.1°
3	384	4.4×10^{-4}	0.2°
4	512	2.8×10^{-4}	0.1°
5	640	2.0×10^{-4}	0.1°

Before this change the joins held 22% of the teapot's nodes on 1.6% of its surface; they now hold 13%. Creases cost less than that: rows within 0.1 of a crease or an end of a sheet are 11% of the body's rows on 5% of its length and 18% of the lid's on 12%. In the baked mesh of Section XI the joins have 1051 of 13 663 vertices; the count there is set by the angle the band turns through, not by its rows.

The interior join was at first built at the wrong place. The spout's channel continues past the body's inner wall into the cavity, and the join was made at the end of that stub, which left a lip standing in the cavity that the 2026 model does not have. The part of a tube to remove is the part nearer the axis than the wall it meets, at both ends: inside the body at the spout's base, inside the cavity at the end of the channel. Cut where it crosses the inner wall and blended to the wall there, the channel opens flush into the cavity. Fig. 2 shows reflection lines running from the body through the join onto the spout without a break.

These joins are not the chamfers of the 2026 model. Their widths (0.11, 0.14 and 0.07) were chosen by eye, and their shape near the handle and spout differs from the official one. What they show is that a join needs no new kind of surface: it is the same blend between two continuous surfaces that CONV already uses between its two factor orders.

The unified teapot has 2912 nodes: 2528 in its four sheets and 384 in the joins (the layout measurements of Section XI were made with the joins sampled, on 3104 nodes), against 5346 in 66, and every sheet lies within 7.5×10^{-4} of its source. (Placed by radius of curvature the same sheets took 5392 nodes and strayed by up to 1.1×10^{-2}.) A 640×400 frame of it takes 44 ms by transport and 56 ms per ray without edge samples.

VIII. How Many Nodes

This section compares node counts. Against the 1976 Bézier patches a CONV lattice is about equal: the patches have about 300 control points, a seamed lattice of 324 nodes is within 2.9×10^{-5} of them, and exact reproduction takes 560. That target is native to Bézier patches. The comparison below gives every method the same points on a surface that is native to none of them. We use three targets: an analytic surface of revolution with a trigonometric profile, half a torus, and the 2026 body with its interior, rows placed as above; and five methods: the nodes joined bilinearly; the Catmull–Rom bicubic [10]; the uniform bicubic B-spline with the nodes as its control points, which on a regular grid is also the Catmull–Clark limit surface [11] of the node mesh; the same B-

spline solved to pass through the nodes, which is the C^2 interpolating spline and needs a global solve along every line; and CONV. Every method is cut at the body's crease.

TABLE VI
Fewest nodes that hold a tolerance on the distance from the target (model units) and on the normal (degrees). A dash: not reached by 6192 nodes (9264 on the body)

	Distance ≤			Normal ≤	
Method	10^{-2}	10^{-3}	10^{-4}	1°	0.1°
Analytic revolution					
Facets	1200	–	–	–	–
Catmull–Rom	156	528	2080	396	2328
B-spline, nodes as control	1584	–	–	–	–
Spline through nodes (global)	78	300	784	150	588
CONV	78	300	784	150	588
Half torus					
Facets	400	4656	–	–	–
Catmull–Rom	54	156	544	108	408
B-spline, nodes as control	600	–	–	582	–
Spline through nodes (global)	42	72	272	54	204
CONV	42	72	204	54	156
2026 body with interior					
Facets	4656	–	–	–	–
Catmull–Rom	588	2328	9264	–	–
B-spline, nodes as control	4656	–	–	–	–
Spline through nodes (global)	294	1040	4632	2316	–
CONV	392	1164	3088	2316	–

Table VI is read from a grid of lattice sizes, so its entries are steps of that grid. On the smooth targets CONV needs exactly what the C^2 spline needs, or less, and a half to a quarter of what Catmull–Rom needs; facets and the raw control mesh are one to two orders behind or never arrive. The spline reaches this by solving a banded system along every line, so each node influences the whole surface; CONV reaches it from the 6 by 6 nodes around a cell. On the body all the cubic methods are within a grid step of one another, because the count is set by the three tight bends and not by the method. CONV is the only one of the five whose normal is not exactly continuous where its limiter acts: at the coarsest sizes it shows kinks of a few degrees, which vanish as the lattice resolves the shape (none above 0.03° on the torus at 272 nodes). The result is equality with the C^2 interpolating spline at local cost. CONV does not need fewer nodes than it.

IX. Against the Later Teapots

Each later version fixed a named defect. Table VII lists what a node lattice does about each; the measurements follow.

TABLE VII
Defects addressed by later versions, and by a CONV lattice

Defect of the 1976 patches	Fixed in	CONV lattice
Body not round	1992	Improved: 233 times rounder at the same node count
Kinked knob tip	mesh options	Improved: the kink is removed
No bottom	1987	Flat cap with an exact crease; Crow's round bottom from samples
Curvature jumps at patch joints	2026	Per patch: not reproduced. Unified sheets: mean relative jump 0.003
Handle and spout pierce the body	1992, 2026	Blended joins of our own (Section VII), not the 2026 chamfers
No interior	1992–2026	The 2026 interior, from samples
Gap around the lid	1992	Not addressed
Mesh cracks, duplicate vertices, pole triangles	mesh options	Not applicable: no mesh is made
Texture layout	2026	One chart per part; patch coordinates also offered

A. Roundness

A cubic Bézier arc is not a circle, and the teapot's quarter arcs use a rounded handle length. At its widest ring the original body is out of round by $(r_{\max}-r_{\min})/\bar{r} = 4.1 \times 10^{-3}$. A node lattice needs no new patch type to do better: put the ring's nodes on the circle. A seamless periodic CONV ring through them is not a circle either, but its error falls quickly with the node count (Table VIII). With the 16 nodes per turn of the exact lattice it is 233 times rounder than the original, and the body and lid stay within 4.3×10^{-5} of the exactly revolved profile. Because the profile factor is linear in the radius, the profile itself remains exact in this style.

TABLE VIII
Out-of-roundness of the body at its widest ring

Surface	$(r_{\max} - r_{\min})/\bar{r}$
Bézier original, and CONV exact	4.1×10^{-3}
CONV round, 12 nodes per turn	9.6×10^{-5}
CONV round, 16 nodes per turn	1.7×10^{-5}
CONV round, 24 nodes per turn	1.6×10^{-6}
CONV round, 32 nodes per turn	2.8×10^{-7}

The same change repairs the knob tip. The original lid ends in a ring of radius 0.01 whose four quarter patches meet at a kink: the tangent around the ring turns by 18.9° a seventh of a node interval below the tip and by more nearer to it. Meshing options exist on the official page for such tips. On the round lattice the ring is a circle and the turn is zero.

B. Curvature continuity

Curvature continuity is the contribution of the 2026 model [6]: where two patches of the original meet with matching tangents but different curvature, a reflection bends. Fig. 3 repeats that test on our lattices. The 1976 lattice, being the 1976 surface, shows the bend at the joint

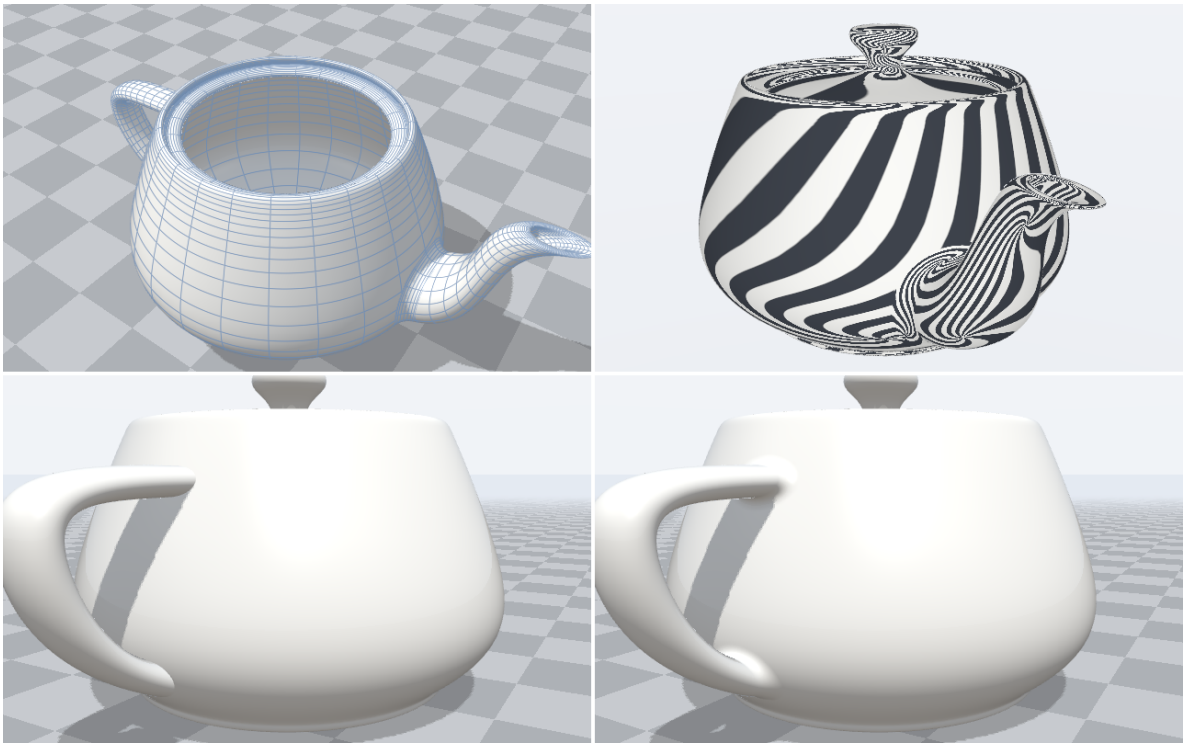


Fig. 2. The 2026 teapot as four sheets with blended joins. Top: the body sheet with the lid removed, its rows gathering at the rim and the foot and continuing down the interior; reflection lines crossing the join at the base of the spout. Bottom: the handle without joins and with them.

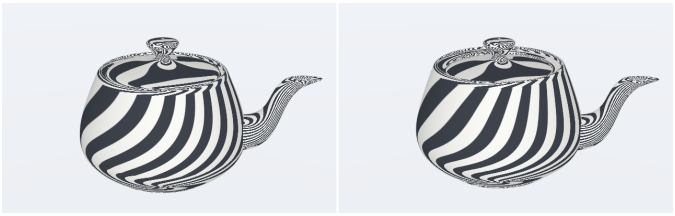


Fig. 3. Reflection lines. Left: the 1976 lattice; the stripes bend where the upper and lower body meet. Right: the 2026 lattice from samples.

of the upper and lower body. The 2026 lattice shows none at this scale.

The figure does not show what the measurements show. At every joint between two exterior lattices we compared the two surfaces' normals and the curvature of the curve that crosses the joint (Table IX). The sampled 2026 lattices keep the normal to 0.11° on average and 0.83° at worst. They do not keep the curvature: its relative jump across a joint averages 0.16, about what the 1976 patches have by design. The reason is the seam. Each lattice ends with one-sided stencils, whose second derivative is accurate to second order only, so two lattices that sample one G^2 surface still disagree about its curvature at their common edge. Our 2026 teapot therefore has the shape of Yuksel's, to the accuracy below, and approximately its normals, but it is not itself curvature continuous. Lattices that run across the smooth joints without a seam would be, where the limiter is inactive. Section VII builds them, with rows placed by a projective warp, and the mean relative jump falls to 0.003.

TABLE IX
Across the joints of exterior lattices, 9 by 9 nodes each

	Normal turn (deg)		Curvature jump	
Lattices	mean	max	max	mean rel.
1976, 44 joints	0.06	1.71	2.85	0.19
2026, 52 joints	0.11	0.83	49.4	0.16

The maximum for 1976 is the kink at the knob tip, and the large maximum for 2026 is on the narrow tubes, where curvature itself is large. Accuracy of the sampled shape. We have no finer truth for the 2026 surfaces than the samples, so we hold some out: lattices built from every other sample, 5×5 per patch, predict the omitted samples to 2.1×10^{-3} at worst and 6.6×10^{-4} rms on the exterior (3.7×10^{-3} and 8.0×10^{-4} on the interior), on a body 4.0 wide. The 9×9 lattices have half that spacing and fourth-order ends, so their error should be near a sixteenth of these figures; that is an expectation, not a measurement.

For the 1976 data alone: around the body and lid, the round style has no seams, and one-sided finite differences find no curvature jump across node lines (relative jump below 2×10^{-8}). Along the profile, the exact and round styles copy the original and therefore its curvature jumps: the worst is 2.85 in absolute curvature, at the joint of the rim and the upper body. They are not curvature continuous there, and Yuksel's construction [6] is.

One might hope to obtain continuity by removing the seams along the profile and leaving them only at true creases. We built that variant, with rows graded in arc

length so that spacing changes gradually. It is not a success at the node counts of interest. At four intervals per patch it deviates from the original by up to 4.0×10^{-2}, and it is not smooth either: where the tight curl of the rim falls between nodes, admission acts and leaves tangent turns of up to 13° at node rows, and the cap crease is rounded to 47° where it should be 90°. Doubling the nodes brings the deviation to about 2×10^{-2}. A CONV lattice is smooth where its data are resolved and is not where they are not. We recommend the seamed styles.

C. What is not addressed

In the per-patch versions the handle and spout pass through the body. The trimmed and chamfered surfaces of the 1992 to 2026 models are not grids of samples, and we do not reproduce them; the unified version replaces them with joins of its own (Section VII). The 1992 and 2006 versions are not included. In the 1975 to 1987 versions the lid still floats 0.1 above the rim.

D. No mesh

Several options of the official page concern tessellation: duplicate vertices along patch edges, triangles at the tips, matching normals across edges, the direction of triangulation. A node lattice has one node where patches meet, and Section X renders the surface itself. When a mesh is wanted for speed, Section XI makes one from the lattice, with one vertex where pieces meet and two only along a crease. A pole is handled by the solver's damped step and by a normal averaged around the pole a short distance along v.

X. Rays by Transport

A camera ray meets $S(u, v)$ where two scalar equations vanish. The usual method searches for that point from the ray's side [13], [14]: a hierarchy of bounds, then Newton's method. Correspondence transport, introduced for a cloth experiment built on CONV, works from the surface's side. The parameters of a lattice of surface points are known; where those points land on the screen tells nearby pixels where their rays must meet the surface.

A. Method

1) Transport. Evaluate each part on a lattice of k samples per node interval ($k = 4$) and project the points.
2) Predict. Each projected lattice cell, split into two triangles and dilated by 0.6 pixel, gives every pixel centre it covers a candidate (u, v) by barycentric interpolation, with a lower bound on depth. A pixel keeps up to six candidates in depth order. Workers own disjoint bands of rows, so the lists need no locks and their order is deterministic.
3) Correct. In depth order, a pixel refines a candidate on the true surface by Newton's method with the dual-number derivatives, to 0.01 pixel. It stops at the first converged hit that is nearer than the next candidate's bound.

No ray performs a search. The triangles are predictors only; every hit, normal and texture coordinate comes from the surface.

Two refinements carry most of the gain. A prediction whose first residual is under 0.35 pixel takes one exact Newton step and is accepted without a second evaluation, provided the step stays inside one cell and is not long for its residual, which excludes creases and grazing rays; the error of such a step is of second order in the residual. And a solve that fails to halve its residual after two iterations is abandoned; without this, rays that just miss a silhouette cost eight evaluations each and dominate the frame.

B. Comparison

The baseline traces each ray independently through a bounding-volume hierarchy over the same lattice cells, intersects the same two triangles per cell, and corrects with the same solver and the same one-step rule. It is a strong baseline: it differs from transport only in how a ray finds its candidates. Table X gives medians of 21 frames of the exact teapot on an Apple A18 Pro (six cores), Apple clang 21 at -O3 without contraction. The machine was shared during the measurements (load average near 6), so the absolute times are indicative and the ratios are the result.

TABLE X
Frame time (ms), one sample per pixel

Frame	Threads	Transport	Per ray	Ratio
640×400	all	17.4	29.1	1.67
640×400	one	48.4	87.4	1.81
960×600	all	30.3	59.6	1.97
960×600	one	90.9	221.4	2.44

The mechanism is visible in the counters. At 960×600 transport starts 1.09 solves and makes 1.34 surface evaluations per surface pixel; the per-ray tracer starts 1.90 and makes 2.10, and visits 8.5 hierarchy nodes per ray besides. Transport pays 4.8 ms to predict and 4.7 ms to transport the lattice and build the shadow map. A repeat of the last row gave 183 ms for the per-ray tracer, so that ratio should be read as about two.

Agreement. For the 960×600 frame we re-evaluated every reported (u, v): the worst reprojection error is 0.20 pixel and the mean 5×10^{-4} pixel. Against a per-ray reference with a lattice four times finer and no one-step rule, transport agrees on 98,347 of 98,462 surface pixels (99.88%) and the per-ray tracer on 98,349. Nearly all the differences lie on silhouettes and crease edges, where a ray grazes two roots less than a pixel apart and neither method is certain to return the nearer. Neither proves that every intersection has been found.

C. Edges, floor, and gloss

Edges. One sample per pixel leaves stair-stepped silhouettes. A pixel whose neighbour shows another part,

another sheet or no surface receives four more samples on a rotated grid, which reuse the pixel's own predictions. About 2% of pixels qualify, but their rays are the expensive ones, and the frame costs about twice as much: 40.0 ms against 50.4 ms per ray at 640×400 on all threads.

Floor. The view reflected in a planar floor is the view from the eye mirrored in that plane, an identity used for planar mirrors in an earlier photonic transport experiment. Its rays are as coherent as the camera's, so the reflection is one more transport pass from the mirrored eye over the same lattice.

More surfaces. The 2026 version has 66 lattices and 5346 nodes, with interior sheets behind most exterior pixels. A 640×400 frame of it takes 38.6 ms by transport and 55.8 ms per ray without edge samples, and 61.1 ms and 80.3 ms with them.

Gloss. A ray reflected by the curved teapot is not coherent with its neighbours, and transport has nothing to offer it; those rays go through the per-ray tracer, one bounce. The emitter is a rectangle seen along the reflected ray, so a highlight has the shape of a window and not of a point light's lobe. With gloss and a polished floor a 640×400 frame takes 72.9 ms by transport and 96.7 ms per ray, edge samples included.

Wireframe and texture. A hit carries its own (u, v) and their screen gradients, so node lines and patch outlines are drawn in the shader with hidden lines removed for free, and they are curves on the surface, not a cage around it. Textures are small RGB lattices reconstructed by the same CONV evaluator, mapped per part or per patch as on the 1976 journal cover [2].

Fig. 4 shows four renderings and Fig. 5 the solver work per pixel.

D. An audit of the loops

The loops of the CPU renderer were read against a list of what makes a tight kernel fast: bookkeeping paid once per block, each datum found once, the shape fixed before the loop, and work handed out so that no core waits for another. Five changes came of it, each giving the same picture byte for byte on one thread. A surface point carried its value and both derivatives through the admission across the rows; the rows' values do not depend on the parameter across them, whose derivative comes from the basis alone, so the admission now carries two numbers and an evaluation takes 275 ns where it took 374. The predictions were written by 24 bands of rows that each scanned every lattice cell, with six threads taking one chunk each, so that on a processor with two fast cores and four slow ones a slow core set the time; the cells are now filed once under bands of eight rows, which are handed out a few at a time. In the prediction's pixel loop, what is constant along a row or over a triangle is found before it. A trim's loop is tested against its box first. For the unified teapot at 960×600 a frame by transport went from 94 to 60 ms on six threads and from 292 to 184 ms on one, chrome from 161 to 114 ms, and the per-ray method from 142 to 115 ms, on a loaded machine, best of alternated runs (results/cpu_audit.txt). The baked renderer's frame did not change. One defect was found: the band test allowed one row beyond a cell where a prediction can reach two, so a few predictions were lost at each band's edge and the picture depended on the number of threads. Skipping tree boxes behind the tangent plane a mirror ray leaves gained nothing, since the boxes such a ray meets are those round its own origin, and was withdrawn.

XI. Baked Vertices

Tracing evaluates the surface at every pixel. Baking evaluates it once, at vertices placed so that flat triangles between them stay close to it, and from then on a frame costs no evaluation.

A. Where to put the vertices

We tried five layouts on the unified teapot and measured each mesh against its four sheets at seven points in every triangle: the distance from the flat triangle to the nearest point of the surface, found by sliding along the surface from the point with the same parameters and never across a crease, and the angle between the interpolated vertex normal and the surface's normal there. The join bands are left out of the measure: the interior spout join folds on itself, a fault of that join which would be charged to whichever layout met it.

Uniform: every lattice cell cut k by k. Bounds: a tensor grid in which each column and row of cells is cut as often as the lattice data alone require. A chord over a step h of a curve with second derivative at most M strays by at most $Mh^2/8$, and the second difference of the nodes is M in node units, so a cell needs $\sqrt{M/8\varepsilon}$ cuts; no surface evaluation decides it. Tree: every cell is halved, along u, along v or both, until the surface at the middle of each edge and of the cell is within ε of the flat piece, and optionally until the normal turns no more than a set angle along an edge. Neighbours of different depth are joined without cracks by triangulating each piece as a fan that takes in every vertex its neighbours left on its border. Clusters: k-means over dense surface samples in position and normal, run with the mini-batch moving-average updates of e-means [12], each cluster giving its most central sample as a vertex and the dual of the clusters on the sample grid as triangles. The fifth, a family of layouts from the curvature metric, has a subsection of its own below.

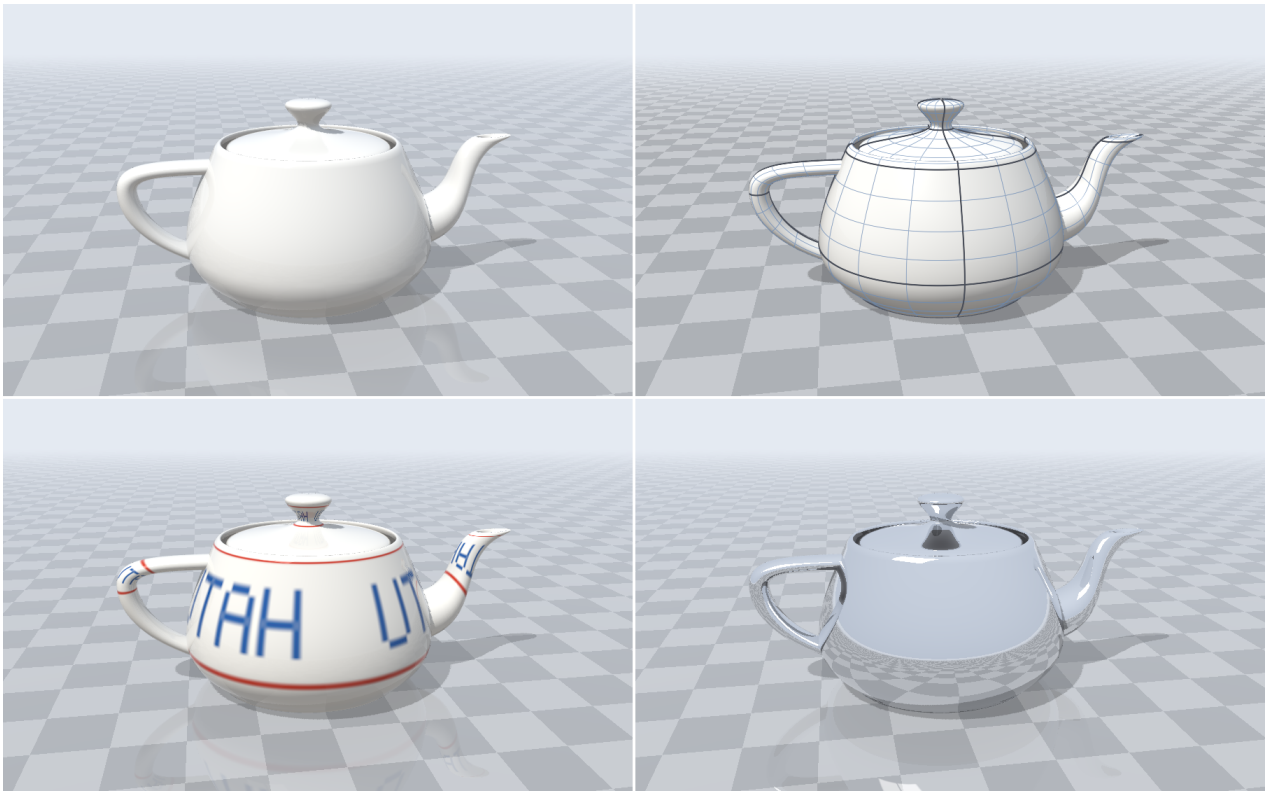


Fig. 4. The CONV teapot, exact style, 560 nodes, traced by transport. Top: porcelain with gloss on a polished floor; node lines (thin) and the original patch outlines (heavy) drawn from the hit parameters. Bottom: a low-resolution texture magnified by CONV under a traced glaze; chrome, reflecting the floor and its own handle and spout.

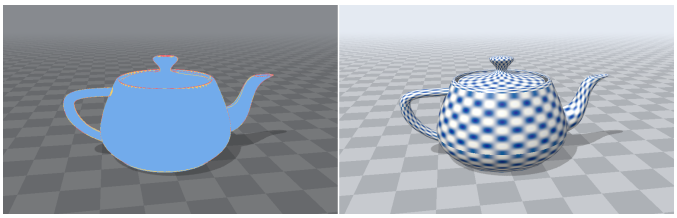


Fig. 5. Left: solver work. Blue pixels needed one solve, orange two, red more. Right: a checker mapped in patch coordinates.

TABLE XI
Vertex layouts on the unified teapot (3104 nodes). Distance in model units; a pixel at the teapot is about 0.011

Layout	Vertices	Distance max	Distance rms	Normal (deg) max	Normal (deg) rms
Uniform 2×2	12786	1.9×10^{-2}	5.1×10^{-3}	39	3.15
Uniform 4×4	49430	4.7×10^{-3}	1.3×10^{-3}	9.4	0.80
Bounds, 10^{-2}	7351	1.4×10^{-2}	3.5×10^{-3}	51	2.21
Bounds, 5×10^{-3}	11060	6.4×10^{-3}	2.2×10^{-3}	51	1.70
Tree, 10^{-2}	10506	9.1×10^{-3}	2.0×10^{-3}	75	1.78
Tree, 2×10^{-2}, 20°	8991	1.5×10^{-2}	4.5×10^{-3}	25	3.86
Tree, 10^{-2}, 20°	14189	9.0×10^{-3}	1.7×10^{-3}	23	1.79
Tree, 5×10^{-3}, 10°	29498	3.9×10^{-3}	1.1×10^{-3}	7.1	1.15
Clusters, 12000	11984	3.7×10^{-2}	1.8×10^{-3}	117	6.58
From the metric: (a) 2×10^{-2}, 20°; (b) 10^{-2}, 20°; (c) 5×10^{-3}, 10°					
(a) nucleated, unchecked	2869	1.6×10^{-1}	8.6×10^{-3}	77	8.61
(b) nucleated	5586	1.9×10^{-2}	4.2×10^{-3}	11	2.69
(b) lattice nodes	6749	1.9×10^{-2}	5.3×10^{-3}	9.9	2.41
(c) nucleated	14144	2.3×10^{-2}	1.7×10^{-3}	58	1.31
(c) lattice nodes	12464	1.0×10^{-2}	2.6×10^{-3}	5.4	1.31

Of the first four layouts (Table XI; the full sweep is in results/mesh_study.txt; the layouts from the metric are the subsection after next) there are three results. First, distance alone is an insufficient criterion. A bend of radius 0.03 fits inside a tolerance of 0.01 as a single flat piece turning 60°, and the shading across it is then wrong however the normals are interpolated: the tree by distance alone has a worst normal error of 75°, and a 20° limit on the turn of the normal along an edge brings that to 23° for a third more vertices. Second, the lattice data do bound vertex placement, and cheaply: the bounds layout holds a distance about as well per vertex as the tree, in half the time and a quarter of the evaluations, but being a tensor grid it cannot follow the bends locally and its worst normal stays at 51°. Third, clustering by nearest centre in position and normal is not competitive: worse on every measure but one, a hundred times slower to build, and its dual triangulation can fold. The uniform 4×4 mesh is the most faithful of the fixed layouts and costs three and a half times the tree's vertices. The picture itself stops improving near 10 000 vertices: against the traced frame, about 500 of 99 000 surface pixels differ in coverage for every layout past that size. The engine's default is the tree with $\varepsilon = 10^{-2}$ and 20°: 14 189 vertices and 26 538 triangles from 49 972 evaluations in 33 ms. A node on a crease becomes two vertices, one per side, each with its side's normal.

B. What sets the count

The problem is a classical one, the best piecewise-linear approximation of a smooth surface, and for flat triangles through points of the surface its answer is known in the limit of many triangles. Locally the surface is $z = \frac{1}{2}(\kappa_1 x^2 + \kappa_2 y^2)$; stretching x and y by $\sqrt{\kappa_1}$ and $\sqrt{\kappa_2}$ makes it a paraboloid of revolution, on which the best triangle is equilateral and errs by half the square of its circumradius. Undoing the stretch, a largest distance ε needs

$$T_\varepsilon = \frac{2}{3\sqrt{3}} \frac{1}{\varepsilon} \int \sqrt{|K|}\, dA \tag{8}$$

triangles, K the Gaussian curvature, each stretched by $\sqrt{\kappa_1/\kappa_2}$ along the direction of least curvature; and a normal that turns by at most α along any edge needs

$$T_\alpha = \frac{4}{\sqrt{3}} \frac{1}{\alpha^2} \int |K|\, dA. \tag{9}$$

Position costs the root of the curvature and smoothness the curvature itself, with different powers of the tolerance. Both integrals can be read from the lattice alone: the five-sample differences that CONV already forms for its jets give the first and second fundamental forms at every node, and from the nodes of the unified teapot $\int \sqrt{|K|}\, dA = 76.4$ and $\int |K|\, dA = 168.5$, against 76.6 and 167.1 from differences of the evaluated surface. The typical ratio κ_1/κ_2 is 15 on the body. By these formulas a distance of 0.02 needs about 740 vertices and a 20° turn about 3200 triangles, where the tree uses 9000 vertices and 26 500 triangles: an order of magnitude, in the limit.

C. Layouts from the metric

To see how much of that can be had, we built the layout the formulas describe. From the nodes, with I, II and $\mathrm{III} = \mathrm{II}\,\mathrm{I}^{-1}\mathrm{II}$ the three fundamental forms in (u, v),

$$M = \frac{|\mathrm{II}|}{8\varepsilon} + \frac{\mathrm{III}}{\alpha^2} + \frac{\mathrm{I}}{L^2} \tag{10}$$

measures a step in units of one allowed edge: its chord strays by ε, or its normal turns by α, or it is L long. Seeds are placed; their cells are grown over a grid of samples by shortest paths in M, so each is one connected piece; the seeds may then be moved toward the means of their cells with the moving-average updates of e-means [12]; and the triangles are the dual of the cells. Creases, ends and poles are pinned rows with seeds of their own. Three ways of placing the seeds were tried: at random; nucleated, one at a time, each where the surface is furthest in M from every seed so far; and at the lattice nodes themselves, whose rows CONV's own error has already crowded where the shape is hard.

Unchecked, the nucleated mesh has the predicted count and is wrong in places by 0.16 (Table XI). Two things that the metric does not contain account for it. A pinned row is a curve in its own right: a chord of the rim strays by the circle's sag whatever the surface's normal curvature along it, so such rows must be sized by the curve's own curvature. And where one principal curvature vanishes, on top of the lip or on a nearly flat disc, the metric calls an edge along that direction free, but a straight chord leaves the line of zero curvature sideways, into surface that bends sharply. That is a third-order effect, absent from the metric. So every triangle is tried against the surface at its middle and the middles of its edges, and one that fails gets a seed where it fails worst.

Checked in this way, nucleated seeds reach the quality of the tree with $\varepsilon = 2 \times 10^{-2}$ and 20° from 5586 vertices in place of 8991, and have no advantage left at 14 000, where a few triangles also remain out of tolerance because the refinement did not converge. To pass the check at the coarse setting nucleated seeds needed 5354 vertices, random ones 6174 and lattice nodes 6550; moving averages brought the random start to 5626 and did nothing for the other two. The lattice nodes give the most uniform result: no triangle is far out of tolerance at either setting, and at the fine one they give a worst normal error of 5.4° from 12 464 vertices, which the tree does not reach with 29 498. Their distance is the weaker side, and the traced picture is still slightly closer to the tree's. Restricting each cell to the lattice cells round its node, and an inserted seed to the lattice cell it lies in, changes the count by 3% and the errors not at all: the metric cells of node seeds already coincide with the lattice cells. The restriction makes each lattice cell independent of the others, so it can be meshed alone. All of these cost about a second and a million evaluations to build, against 30 ms and 50 000. The gain is well below the asymptotic limit. Creases, poles and lines of zero curvature, where the limit's assumptions fail, cover much of the teapot. The engine uses the tree.

D. Rasterizer

The mesh is drawn on the CPU. Edge functions are integers with eight bits below the pixel and a top-left fill rule, so triangles that share an edge neither overlap nor leave a gap; depth is $1/z$, which is affine on the screen; triangles are filed by bands of rows, which threads share out. The rasterizer writes only a visibility buffer, a triangle and two weights per sample at two by two samples per pixel, and shading follows once per pixel, or once per sample where a pixel's samples show different parts or depths. The techniques are those of the software rasterizer in PlaySuite's fish tank; the code is our own.

E. One line where a join meets a part

Along each of its edges a join's band lies on another part's surface, but the two meshes have their own vertices there. Flat pieces of different sizes do not agree on where that surface is or which way it faces, and from some angles the edge showed as a line. Three things remove it. A part is meshed to an eighth of the tolerance and half the angle wherever a band's edge crosses it. Every vertex on a band's edge takes the position and the normal that the other part's mesh has at the same parameters, the point of that part's triangle and its interpolated normal, so the band's edge is a line drawn on the other mesh. And where the

band and the part are both drawn, in the strip the trim leaves under the band, the band is drawn: its vertices are projected 0.15% nearer the eye. The mesh of the unified teapot goes from 13 663 to 16 431 vertices.

F. Light painted by the ray tracer

Each part has a light map in its own chart: around the part one way, arc length along it the other, so texels are spread evenly over the surface however its rows are graded. To bake whole, every texel is shaded exactly as the ray tracer shades a hit at that point seen from the camera, with its shadow, texture and traced reflection, and stored with a chain of half-size levels; a texel facing away from the camera is shaded for the mirrored view direction, so the far side holds its own outward lighting. A frame then reads the map between the two levels that match each triangle's size on screen. 601 808 texels of 0.016 units bake in about 130 ms. A map baked whole holds for one camera and one pose.

G. A bake that goes with the body

The light of this scene has three parts with different dependences. The sky's diffuse light and the sun's cosine factor depend only on the normal as it is turned now, and are closed forms. The surface's own colour belongs to the body. The fraction of the sun that reaches a point past the body itself depends on the direction of the sun in the body's frame, so it changes when the body turns and not when the body or the camera is moved. The map therefore keeps the colour in its three colour channels and the lit fraction in the fourth, and a pixel is

$$\text{colour} \times \big(\text{ambient}(n) + \max(0, n \cdot l)\,\text{lit}\big)$$

with n the interpolated normal carried by the pose. The shadow on the floor is treated the same way: it is baked once as a lit fraction over a square of the floor, kept relative to the body's origin, and looked up at a floor point moved back along the light by however far the body has moved since. A body lifted off the floor leaves its shadow where the light puts it. The drawn floor reads this texture too, so no shadow ray is traced in any frame.

While the body turns, the lit fraction and the floor's shadow are read from the shadow map, drawn from the mesh at 512^2, with no traced edges. When it comes to rest the lit fraction is found again with exact edges. The first bake keeps every texel's point and normal in the body's frame, so this pass evaluates no surface: 207 ms on one thread against 571 for the first bake (about 290 ms against 1090 in the browser). Against the same poses shaded per pixel with no bake, the picture after a shift of the body differs by more than 8 of 255 in 651 of 576 000 pixels, and after a turn of 40° in 385 before the lit fraction is found again and 816 after. In the browser at 960×600 the shading of a moving frame takes 8 ms with the bake and 13 without (frames of 21 and 25 ms), and a frame at rest 51 ms against 65, or 50 against 71 with gloss and the floor's reflection on.

H. What each point sees of the room

The room's diffuse light above is a function of the normal alone, as if nothing were in the way. That is wrong wherever the body is in its own way: the inside of the teapot received the whole sky, darkened by a constant; under the knob, in the gap around the lid and down the spout's channel nothing darkened at all. A second bake, optional and at four times the texel size, stores for every texel and side the fraction of its cosine-weighted hemisphere that 32 rays find open past the body, and the mean open direction in the body's frame. The set is turned by a different angle at each texel and the map is smoothed, so the resolution of 1/32 shows neither as bands nor as grain. The room's light at a pixel is then the open fraction times the room's light from the mean open direction as the pose turns it. This holds for any motion of the body.

With that alone the lid, the knob and the channel are shaded by their own geometry, and the interior is nearly black: most of what reaches the inside of a white pot has bounced off the pot. Two ways of adding that were tried. One value for the whole cavity, the light entering it divided by what a bounce loses, makes the interior bright and uniform and the channel as bright as the cavity. The other keeps, for every texel, the texels on which its blocked rays land. This transfer belongs to the body. Whenever the sun or the room turns relative to the body, the direct light E_0 of every texel is found (room light through the open fraction, and the sun by the shadow map) and $E = E_0 + \rho\, T E$ is solved by six gathers through the transfer with albedo $\rho = 0.9$; $E - E_0$ is stored and added at each pixel. The interior then has the level a white cavity in sun should have, and the channel and the knob's underside receive what their surroundings throw on them. The gather is part of the pass that finds the lit fraction again and that pass still takes about 190 ms. The bake is where the cost is, and three things brought it down. Of 32 directions a quarter are tried first and the rest only if those differ. Every second texel of every second row is done first, and a texel between four that see the whole room is taken to see it too, as is, on a closed solid, a texel between four that see nothing (the side of a sheet that is inside the material). And a part whose one side is found by trial to be wholly inside the material has nothing baked for that side. In the browser the bake takes 0.7 s without the open fraction, 1.1 s with it and 1.3 s with the transfer (0.7, 1.9 and 3.3 s before), and the baked renderer now uses the transfer by default. A frame at rest costs 43, 46 and 48 ms. While the body turns the gathered light is not refreshed; it turns with the body until the body rests. The option to leave the surface's pattern off whatever faces the cavity is separate. These modes exist only in the baked renderer; the ray-traced renderers still use the normal alone.

I. Reflections from emitters

A light map baked whole freezes whatever depends on the view: turn the camera and the highlights and the

floor's reflection stay painted where they were (Fig. 6, right). With the map holding only the colour and the lit fraction, what depends on the view is added at every frame. Each pixel has an interpolated normal, hence a mirror direction, and that direction is answered by emitters and not by tracing. Above the horizon it is the sky and its window. Below, it is a stand-in for the floor: an object that is never drawn, whose pattern is the floor's in closed form at the point the mirror ray reaches, and whose shadow was baked once into a small texture with the exact shadow edges described below. The drawn floor and the emitter are separate things that agree. The result follows the camera at the cost of a few arithmetic operations per pixel: 11 ms for a 960×600 chrome frame against 141 ms traced (34 against 483 on one thread). The emitters do not include the teapot itself, so the knob is not reflected in the lid nor the handle in the body. For the body, the handle and the spout (the lid is left out for now), the mirror ray is now met with the baked mesh's own triangles and followed from piece to piece up to three times: the spout in the body, the body in that spout, the spout in that body. Each point it reaches is shaded from its light map, and the last one's mirror is answered by the emitters, so the floor deep in the reflections is the same stand-in, with the same baked shadow, as in the first. Where the body has moved, the floor is met where it now lies. The triangles stand in a tree of boxes, one subtree a part, built with the mesh and carried by the pose when the body moves. The routes between the pieces are baked with it. For every triangle and every one of 256 cells of directions, the bake marks whether a mirror ray from the triangle in that cell may land on a piece at all: the triangle's possible starting points lie in a sphere, each box of the tree in another, and the cell in a cone, and a cell is marked only if its cone meets the angle some box's sphere fills, for boxes in front of the triangle's tangent planes. A ray in an unmarked cell leaves the pieces at once and is not walked; the picture is the same with the routes as without them, pixel for pixel. The bake takes 0.12 s on six threads (0.41 s on one). A chrome frame at 960×600 takes 29 ms with three bounces, against 34 ms for one bounce walked without routes and 21 ms with the emitters alone. The routes mark which directions may land, not where: storing the landing point too, so that a bounce is a lookup instead of a walk, is left for later.

J. The mesh as a seed

The mesh also serves the ray tracer. Each visible sample names a triangle and weights, hence a parameter pair within the mesh's tolerance of where the ray meets the surface, and one Newton solve from there lands on the surface itself. Transport gathers up to six predictions per pixel and solves several; a seeded ray solves one. Where the mesh catches a ray that the surface does not, which happens within the tolerance of a silhouette, the mesh's own point is shown.

TABLE XII
One 960×600 frame of the unified teapot, medians of 9

Method	Time (ms), threads all	Time (ms), threads one	Evaluations per pixel
Per ray, hierarchy	150	410	4.1
Transport	91	261	3.8
Seeded by the mesh	37	125	2.0
Mesh, shaded per pixel	15	45	0.16
Mesh, baked light, emitters	11	36	0.08

The evaluations left in the last two rows of Table XII, and 0.6 of those in the row above, are shadow rays (next subsection). In a browser, on one thread in WebAssembly, the same frame takes about 450 ms by transport, 200 ms seeded and 40 ms from the mesh with baked light; traced chrome takes 750 ms there and the mesh with emitters 50.

K. Shadow edges on the surface

Shadows are looked up in a depth image drawn from the light. Its texels are larger than a pixel, and a shadow's edge followed them in stair-steps. Blending the lookup only rounds the steps. The edge belongs to the surface, so that is where we now find it: each depth texel also records which part it saw and where, and at a point whose surrounding texels disagree the light ray is traced by Newton from those records, two rays on a rotated grid inside the pixel and two more if they differ. A texel that saw the receiving surface's own neighbourhood is not counted as disagreement, which keeps the wide band of doubt along a terminator from being traced. The edge is then as sharp as the surface makes it at any zoom, for 0.1 to 0.6 evaluations per surface pixel.

Where the light grazes the surface a traced ray meets the surface's own neighbourhood at a glancing angle, so the traced answer is weighted by how squarely the surface faces the light and the depth image's answer takes over toward the terminator. That is wrong where another part's shadow falls across a terminator, as the body's does across the base of the spout: the depth image's one-texel ramp is drawn out by the grazing angle into streaks. When a texel near the point holds something more than 0.3 nearer the light, the traced ray is used in full; it meets that occluder squarely. The streaks go, and a frame at rest costs 4% more. In the light map the same edge is limited by the texel, 0.016 units, which shows as steps when the camera is flown close.

L. Reflections of the floor, and work that is skipped

A curved mirror shows the floor's checker much reduced, and a point sample of it aliases: the lower body of a glazed teapot showed a speckle that looked like facets. The checker is the product of two square waves, and each can be averaged exactly over an interval, so a mirror ray is answered with the checker averaged over the square its pixel covers on the floor. The side of that square is the distance between the point where this pixel's mirror ray meets the floor and the points where the rays of the pixel to its left and the pixel above did.

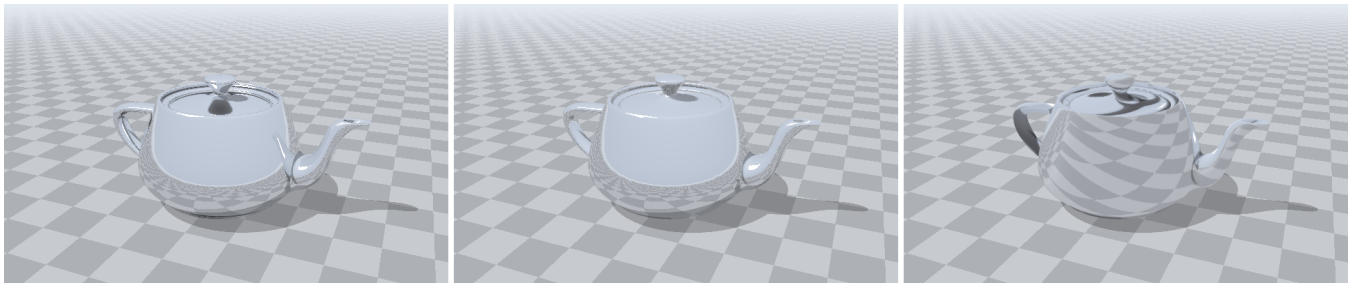


Fig. 6. Chrome. Left: traced, one bounce. Middle: the baked mesh, its reflections taken per frame from emitters, the sky above and a stand-in floor below; only the teapot's reflection of itself is missing. Right: the same mesh with reflections baked from another viewpoint and left there.

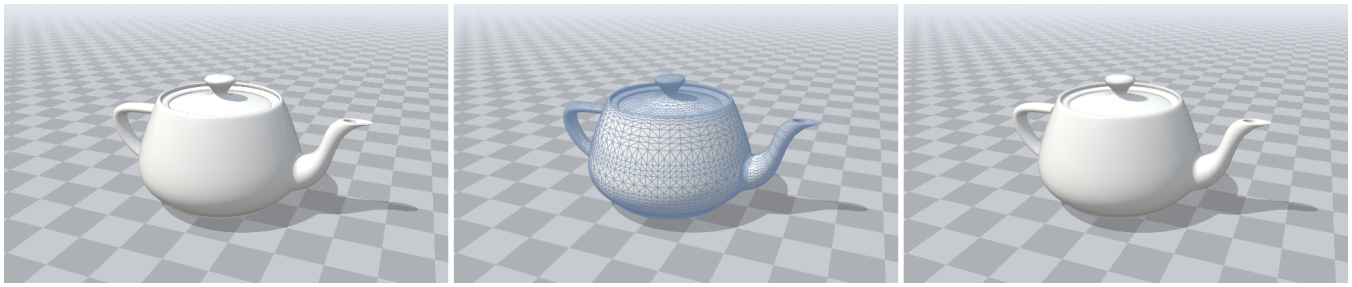


Fig. 7. Left: traced by transport. Middle: the baked mesh, 14 189 vertices, with its triangles drawn. Right: the same mesh lit from the light map painted by the ray tracer.

Where no ray is traced, an emitter is not in view from everywhere. A point on a surface that faces the teapot's cavity (the body's inside, the lid's underside, the spout's channel) cannot see the floor through the bottom. Its mirror ray is given the floor or the sky only if it leaves through an opening, the cut of a cutaway or the top when the lid is off, tested in the teapot's own frame so that the answer is the same when the teapot is turned over; otherwise it is given a dark neutral value for the cavity. The traced modes need no such rule.

The parts of the sheets that lie under a join, the end of a tube inside the body and the body under a band, are not surface. Their lattice cells are left out of prediction, of the shadow map and of the tree for incoherent rays, so no ray is solved against them. The lattice points, the shadow map and that tree depend on the surfaces and not on the camera, and are kept from frame to frame while the surfaces stay where they are: 23 ms of a frame without gloss and 33 ms with it, on one thread. The times of Table XII were taken before this.

M. Cutaway

The official page shows the left or right half of the model. We cut the teapot along its plane of symmetry and fill the cut where it passes through material. A ray that starts in the removed half and has crossed the surface an odd number of times there is inside material when it reaches the plane, and shows the cut face; otherwise it continues to the first surface in the kept half. The count is made by drawing the surface in the removed half as triangles and counting, at each sample, how many cover it: the baked mesh's triangles, or for the ray tracers the cells of the transported lattice. Counting the hits among a pixel's solved predictions was tried first and is not reliable, because the list is short and a missed or repeated hit changes the answer along every silhouette. Lattice cells wholly in the removed half are also left out of prediction, so they do not displace the cells that are seen. The count requires every point of the teapot to belong to exactly one sheet. The joins overlap the surfaces they join by a few percent of their width so that no gap shows, and in that overlap a ray would be counted twice; the count therefore uses a second set of trim boundaries placed exactly at the bands' edges. What remains is a mismatch one sample wide between those boundaries and the bands, which is removed from the count. Geometry in the removed half casts no shadow, so light reaches the interior.

XII. Motion

Since every node is a surface point, the collision shape of the teapot can be taken from the lattice with no separate model: the convex hull of a group of nodes is a convex piece of the teapot. For the 1975 to 1987 versions we use eleven hulls, one for the body, two for the lid, and four each along the handle and spout; for the 2026 version, one hull per group of exterior lattices. A hull is not taken through the nodes of a ring: with 12 nodes to a ring that hull lies inside the surface by 3% of the radius, and a teapot resting on its side showed the floor through its wall. Each ring gives the hull at least 16 points of the surface (20 on the body and lid), moved out from the ring's centre so that the hull's flat sides touch the ring; a round part gives the 16 rings of its profile's convex outline, sampled between rows as well, that leave the least of the outline

outside. Dropped at tilts from upright to upside down, the resting surface is then within 0.007 of the floor in the unified version and 0.001 in the 1976 version, where it had been 0.015 and 0.018 into it. The renderers let the floor hide the surface only where the surface is more than 0.012 below it. We hand the hulls to the rigid-body engine of a rock-stacking game, whose wrench-transport solver resolves contact, friction and rolling resistance in one solve per frame. One model unit is taken as 7 cm. The engine returns a pose. The lattice stays at rest and the pose is applied to the surface: every evaluated point and derivative is turned and shifted. An earlier version applied the pose to the nodes and built CONV through the moved nodes. That is a different surface. Admission and the bubble act on each coordinate separately, so they do not commute with a rotation, and a monotone coordinate of the lattice at rest need not be monotone after a turn. Table XIII measures the difference for a turn about one axis: none at 90°, where the coordinates are only exchanged, and otherwise up to 1.3×10^{-2} in position and 62° in the normal for the 1976 lattice. The moved-node surface shows thin creases under grazing light that the lattice at rest does not have. The same holds for any rigid motion of a CONV surface whose admission is not inactive in every frame.

TABLE XIII
CONV through turned nodes against the turned CONV surface

		Distance		Normal (deg)	
Version	Turn	max	rms	max	rms
1976, exact	30°	1.3×10^{-2}	1.0×10^{-3}	62	1.91
1976, exact	60°	1.2×10^{-2}	8.4×10^{-4}	62	1.90
1976, exact	90°	1.3×10^{-15}	1.3×10^{-16}	0	0
2026, unified	30°	1.2×10^{-3}	3.3×10^{-5}	26	0.54
2026, unified	60°	1.0×10^{-3}	2.7×10^{-5}	23	0.40
2026, unified	90°	1.3×10^{-15}	1.1×10^{-16}	0	0

Dropped from 42 cm at a tilt of 1.2 rad, the teapot lands on its spout, rocks back over its base and is asleep within 1.5 s (Fig. 8). The body is treated as solid porcelain. A real teapot is a shell; simulating one would use the interior, which these versions lack.

Taking hold of the teapot is a pin at the point that was picked: a critically damped spring of 8 Hz between that point of the body and the pointer, applied as an implicit impulse at the point each engine frame, with the engine's contacts solved after it. Lifted by its handle the teapot turns through 90° and hangs from the handle, its centre below the hand; it is not carried level by its centre of mass.

XIII. On the GPU

A second version of the demonstration draws and simulates the same teapot with WebGL 2. The CPU makes the teapot, that is, places its nodes, and hands the nodes over as textures. Everything after that is computed in shaders: the surface, the mesh it is drawn from, the tree its rays walk, its light, and its motion.

A. CONV in a shader

The engine hands over the lattice nodes and the shape of every cell (which difference stencil each node takes, decided by its place in the line). A shader admits the u-cells of every row from the nodes, once per teapot, into a texture: seven numbers per coordinate per cell. The v-cell through the eight row values at the query has to be admitted at the query, because its inputs depend on u. Admission is piecewise linear in its inputs: every step is a sum, a multiple by a constant, or a choice between two values by comparing them. So the derivative along u is carried through it as the second component of a two-vector, taking at each choice the derivative of the value chosen, which is what the CPU's dual numbers do; the derivative along v is the derivative of the basis. A band between two parts evaluates its two parameter lattices, then the body and the tube there, and blends them. The surface is evaluated once at every point of a grid over each part, the grid's steps dividing the node intervals evenly, into a texture; there is no vertex buffer, a vertex's number saying which cell and corner of the grid it is, and each frame carries the points by the pose. It is evaluated again at every pixel for the normal. Numbers are 32-bit floats. At 12 000 points spread over the parts the shader's surface is within 2.3×10^{-6} of the CPU's and its normal within 0.027°. The Bézier patches and the control mesh are evaluated in the shader too, within 1.1×10^{-6} and 4×10^{-7} of the CPU's, and a texture's lattice of colours by CONV, bilinearly or by the nearest node.

B. The picture

Lighting, the shadow map, the trims of the joins, emitter reflections and the polished floor are the CPU renderer's formulas. A cutaway counts crossings of the kept half in the stencil buffer, and an odd count shows the cut face. The teapot is drawn first for depth alone and then shaded at the nearest surface of each pixel, so the surface is evaluated once per pixel and not once per layer behind it.

Two things the CPU does with rays are done with rays here too. Where the shadow map's texels around a point disagree, light rays are traced from a rotated grid inside the pixel, as on the CPU; and a glossy teapot's mirror ray is traced one bounce, the point it reaches being shaded as the CPU shades a surface seen at second hand. The rays meet the mesh's triangles, not the surface itself.

The mesh is the bounded form of the layouts from the metric (Section XI-C): the seeds are the lattice's own nodes and each lattice cell is meshed alone. The surface is sampled once, eight places to a cell each way. One fragment a cell reads its two edges and its middle line each way and finds the fewest equal steps, from 1 to 16, for which a chord stays within ε of the surface and the normal turns by no more than α along it: with P'' and N' taken from the eight pieces, $k = \max(\sqrt{|P''|/8\varepsilon}, |N'|/\alpha)$ where they are largest. An edge's count depends on the edge alone, so the two cells that share it agree; a cell's

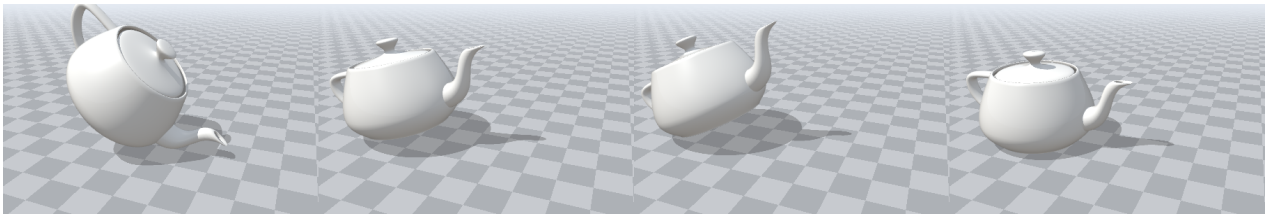


Fig. 8. A drop, at 0.30, 0.42, 0.60 and 1.60 s. The surface is drawn from its lattice at rest, carried by the body's pose.

inside takes the largest count each way, and its border vertices are placed at the edge's own steps, so neighbours meet. Then one fragment a triangle corner evaluates the surface. With $\varepsilon = 0.003$ and $\alpha = 8°$ the unified teapot is 73 560 triangles, where the grid used before had 132 096.

Over the mesh stands a tree of boxes written in the order a ray walks it. A node is one texel of four whole numbers: its box in 65 535ths of the teapot's box, rounded outward, and the node to go on from if the ray misses it. A node the ray meets is followed by its first child, which is the next node. So a step is one read, and the ray keeps no stack. The tree halves a part's cells by where they are in space, down to one cell, and goes on halving the cell's steps down to one pair of triangles. Two shortcuts for a mirror ray are exact. The ray stays in front of the tangent plane it leaves, so a box wholly behind that plane is skipped. And each cell is told beforehand which of sixteen blocks of its own part lie wholly behind the tangent planes at nine places of the cell; a ray from the cell skips those blocks, which for most of the body's outside is the whole body. A step takes about a nanosecond. Most mirror rays then take fewer than 16 steps; rays from the lid, which is hollow round its knob, and from the inside of the handle take 64 to 255, and are most of what tracing still costs.

The normal at a pixel is read from a map of the surface's normals laid over each part. The shadow map is drawn again only when the teapot, its pose or the sun changes. The picture's shader is compiled for the settings in use, each setting a constant in it, so that the work of a pixel holds no decision that was made before the frame began.

Against the CPU ray tracer at 960×600, the share of pixels that differ by more than 8 of 255 is 0.25% for porcelain, 0.25% with the polished floor, 0.39% with a texture, 0.55% for chrome and 0.96% for a cutaway. Without the traced mirror ray chrome differed in 3.6%. Those figures are for the grid and the pyramid of boxes used first; against that build the present one differs in 0.15% to 0.6% of pixels in seven views, along edges, and in 1.65% seen from above in chrome, where the spout shows the body's rim and the lid's knob, which the pyramid's walk missed. A frame takes 3.8 ms at 960×600 at the page's settings, in a headless browser on the laptop used throughout (10 ms with the grid and pyramid). At 2560×1600 with chrome, the polished floor, traced rays and a shadow map of 4096^2 it takes 65 ms (111 before), and 15 ms without traced rays (45 before). The CPU's baked renderer takes 21 ms for a moving frame and 43 to 50 ms for a resting one at 960×600.

C. Room light without rays

The CPU bake of Section XI-G casts rays from every texel. A rasterizer is better at the opposite order. The body is drawn from each of 256 directions spread over the sphere, keeping the distance along the direction of what is nearest; every texel of a map laid over the parts then looks itself up in each picture and, if nothing is nearer, adds the direction weighted by its cosine. The sum is the open fraction and the mean open direction. This takes about 30 ms. For the light the body throws on itself the CPU stores which texels each texel's blocked rays land on. Here that is found again each time: along each of 48 directions the body is peeled into six depth layers, each layer carrying the light of its far side, and a texel that faces the direction with the body in front of it takes the layer just in front of it; five such gathers take 0.7 to 1.0 s when the teapot comes to rest in a new attitude. Against the CPU's baked renderer the mean difference of the picture is 1 to 2 of 255; the two estimate the same integrals from different samples.

D. The rigid body

The CPU version's rigid-body engine (Section XII) solves a frame's contacts together: each contact point keeps the impulse it carried last, and a frame runs a few passes, each minimising over the body's twist a convex cost by Newton's method with an exact line search. That solver is ported to a fragment shader. One run of the shader is one frame. The contacts are the hull vertices within the margin of the floor; the frame's rows live in arrays inside the shader; the Newton system is 6×6, since there is one body, and is solved by elimination; the line search is the CPU's. A fragment can write only its own texel, so the body's new state, a few texels, is written by fragments that each run the frame, and the impulses the contact points retain are a pair of texels per hull vertex. A vertex's impulse after a pass depends only on its impulse before the pass and the body's twist after it, so the solve also writes the twist after each pass and each vertex in contact follows its own row through the passes from those twists, without solving again. Numbers are 32-bit floats, and Newton stops at a relative gradient of 10^{-5} where the CPU stops at 10^{-9}.

The two were run side by side, frame by frame. Over a drop of 600 frames the pose's position differs by at most 2.6×10^{-3} model units for the 2026 teapot and 3×10^{-6} for the 1976 one (the teapot is about 3 units high); over

a grab at the handle, dragged and released, 360 frames, by 8×10^{-5}; over a poke, by 2×10^{-5} to 4×10^{-5}. The state is written through four render targets, so that six fragments run the frame. Both come to rest in the same pose to 3×10^{-4}. A frame of the shader's solver takes 1.4 to 2.4 ms with the read-back of the state, against 0.13 ms on the CPU. An earlier, simpler step (a mean over the contact points of the impulse each asks for, with contact points found on the GPU as the farthest points of the surface in each direction from the centre of mass) is kept as an option; it reaches the same resting pose for the drop within 0.16 units but is not the same solver.

The body's mass, centre of mass and inertia are computed by a shader that sums signed tetrahedra over the faces of the collision hulls. They agree with the engine's to 8×10^{-7} (mass, relative), 2×10^{-7} m (centre) and 4×10^{-6} (inertia, relative).

E. Placing the rows

The rows of the unified teapot's body and lid are placed by CONV's own error (Section VII): the wanted spacing at an arc length is $(e/h^4)^{-1/4}$ for a cell of length h whose CONV line lies e from the source meridian, a projective grading turns that spacing into rows, and the step is repeated, keeping the plan with the smallest error. On the GPU each step is a pass over a small texture. The engine hands over the meridian of the sampled patches as a table. One pass writes the wanted spacing at each station of each section between creases; one texel finds, by 120 steps of bisection, the scale for which the gradings hold the asked number of rows; one pass writes each row's arc length; one pass admits, for each cell, the two CONV lines through the rows and measures their distance from the meridian at seven parameters; one pass keeps the better plan. Nine rounds take about 0.1 s a sheet. The engine then builds its lattice on the rows handed back, and a shader writes the nodes on those rows.

The first plan, which uses the meridian's radius of curvature, agrees with the CPU's to 2×10^{-4} to 7×10^{-4} in arc length on meridians 8.5 long. The kept plans have the same number of rows (73 and 55) and nearly the same error, measured on the CPU in 64-bit for both: body 6.95×10^{-4} on the GPU's rows against 7.03×10^{-4} on the CPU's, lid 2.53×10^{-4} against 2.46×10^{-4}; at the taller height scale, body 6.62×10^{-4} against 5.79×10^{-4} and lid 2.25×10^{-4} against 2.33×10^{-4}. The rows themselves are up to 0.16 apart in arc length (0.31 at the taller scale): the rounds produce several plans whose errors lie within a few percent of each other, the selections in the grading (minimum over neighbours, growth limits) amplify the difference between 32-bit and 64-bit arithmetic, and a different plan is kept. The nodes the shader writes are within 6×10^{-7} of the engine's on the same rows, and the surface check of the shaders against the CPU, both on the GPU's rows, stays at 2×10^{-6}. The rows of the handle and the spout are placed by the same passes on each tube's seam (25 and 37 rows, errors 1.1×10^{-4} and 2.9×10^{-4}).

F. Building the teapot

The engine asks the page for each piece of the teapot's construction at the point where it would compute it, and uses what comes back. The nodes of the 1975, 1976 and 1987 lattices are sampled from the Bézier patches by a shader. The nodes of the unified body and lid are the meridian at each row, turned about the axis; those of the handle and spout are points of the source patches' CONV surfaces around each row. For a join, one pass finds for each column of the tube where it crosses the body's wall: a search along the tube in quarter rows and then bisection, where each test is itself a bisection on the body for the wall's radius at that height. A second pass walks the band's two sides from the crossing, on the body away from the loop's centre and along the tube, in eight steps that read each surface's speed, and writes the two parameter lattices and the trims. The rings of surface points the collision hulls are made from are evaluated by a shader as well. The engine keeps the bookkeeping: chaining the source patches into a meridian, the shape of each cell, and its own copy of the cells for the checks.

The hulls themselves are made by wrapping. For every ordered pair of points of a set, one fragment turns a plane about the line through the two until no point is outside it. If every point is then on or inside the plane, the two lie on a face; the points on the plane are the face's corners; and the pair is a side of the face if no corner is to its right and none lies between the two. Each side, with the lowest-numbered corner of its face, is a triangle of a fan over the face, and the fans are the hull's surface. The fans close only if every fragment agrees on which points are on a plane. With 32-bit floats they do not: the hulls came out with holes. So the points are put on a grid of whole numbers, 16 000 steps to the set's farthest coordinate, and the side of a plane a point is on is the sign of a determinant computed exactly in 32-bit integers, the normal in three parts of twelve bits. Mass, centre and inertia are then integrated over the fans. Against the engine's hull builder, for the five versions, the mass agrees to 4×10^{-6} to 3×10^{-5}, the centre of mass to 7×10^{-6} m and the inertia to 5×10^{-5}; every corner of the engine's hulls is within 10^{-6} m of one of the GPU's. The GPU keeps a few corners the engine drops when it merges faces within half a degree (895 against 892 for the unified teapot). On these hulls a drop of the unified teapot differs from the CPU's by up to 0.07 units while it bounces and 0.006 at rest; on the engine's hulls the two solvers stay within 2×10^{-3}.

Against the engine building the teapot alone, on the same rows, the nodes agree to 5×10^{-7}, the joins' lattices and trims to 1.1×10^{-5} in parameters, where a row or a column is 1, and the body's mass to 10^{-7}. Building the unified teapot this way takes about 0.4 s.

XIV. Limits

The variation bound is factorwise and coordinatewise; it does not bound the curvature of the surface or forbid self-intersection. Exact reproduction is a statement about a

rest pose whose admission is inactive; a deformed lattice is a CONV surface through the moved nodes, not a deformed Bézier surface. The surface is continuous within a part and only there. Transport and the per-ray tracer both accept a converged root without proving it the nearest, and both use lattice triangles as predictors, so a feature thinner than a lattice cell on screen can be missed. The hierarchy's boxes are padded by a fraction of their diagonal, which is a heuristic and not an enclosure. Timings are of one implementation on one shared machine, and vary by a quarter between sessions. The baked mesh is flat between its vertices: its silhouette is the surface's only to the tolerance it was baked at, a light map holds the diffuse light of one pose, and the emitters that answer its mirror directions do not include the teapot itself. The vertex-count limits quoted are asymptotic and assume elliptic points; their constants are derived here for a paraboloid. The node-count comparison covers surfaces that suit a lattice; it does not cover shapes of arbitrary topology, which subdivision surfaces handle.

XV. Conclusion

A CONV lattice with seams reproduces the 1976 Utah Teapot exactly from 560 points that lie on it. The lattice can be made round by moving nodes onto circles, traced without a mesh, textured in its own parameters, and dropped on a floor by handing its nodes to a physics engine as collision hulls and applying the resulting pose to the surface. Later versions enter as samples, with no knowledge of how they were constructed: the 2026 teapot, interior included, is 66 small lattices. Across their seams those lattices do not carry curvature continuity. Whole sheets reduce the curvature jump from 0.16 to 0.003: the body with its interior is one lattice, the lid another, with rows placed by a projective warp, and the handle and spout meet the body in joins that are blends of the two surfaces. With rows placed by CONV's own error the whole teapot is 2912 nodes, and on targets native to no method a CONV lattice needs as few nodes as the globally solved C^2 spline while reading only the nodes around a cell. When speed matters more than exactness the lattice bakes to a mesh whose vertices its own error places, lit by a texture the ray tracer paints, at a tenth of the cost of tracing. Unfinished: a residue of curvature jump where rows are coarse, and joins that differ from the chamfers Yuksel designed.

Attribution

The Utah Teapot was created by Martin Newell at the University of Utah and scaled by Jim Blinn; the 1987 bottom is Frank Crow's and the 2026 version is Cem Yuksel's. The models built here are Modified Utah Teapots in the sense of the terms on the official page [5]: the 1976 surface re-expressed, with an optional flat cap; and surfaces of the 1987 and 2026 versions rebuilt from points sampled with that page's mesh generator, without trimming or chamfers.

AI Generated Work Disclaimer

Substantial portions of this work were generated with the assistance of Anthropic Claude models.

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